

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 One right-angled triangle

1. A car drives away from the foot of a tower 30 m high. What is the angle of elevation of the top of the tower from the car when it is $10\sqrt{3}$ m from the foot? (a) 30° (b) 45° (c) 90° (d) 60° **CBSE 2026, 1 mark**

SOLUTION

$$\tan \theta = \frac{\text{height}}{\text{distance}} = \frac{30}{10\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\tan 60^\circ = \sqrt{3}$$

Answer: (d)

2. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? **CBSE 2025, 1 mark**

SOLUTION

The height is the side opposite the 30° angle and the string is the hypotenuse.

$$\sin 30^\circ = \frac{150}{\text{string}}, \text{ so string} = \frac{150}{\frac{1}{2}}$$

Answer: 300 m

3. A ladder leaning against a wall makes an angle of 60° with the ground. Its foot is 2.5 m from the wall. Find the length of the ladder. **CBSE 2016, 1 mark**

SOLUTION

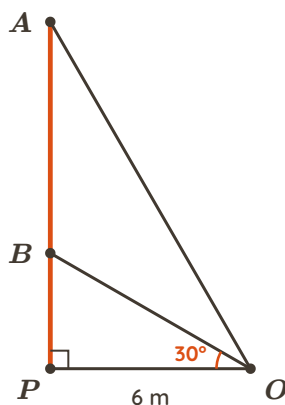
The distance from the wall is the side adjacent to the 60° angle, and the ladder is the hypotenuse.

$$\cos 60^\circ = \frac{2.5}{\text{ladder}}, \text{ so ladder} = \frac{2.5}{\frac{1}{2}}$$

Answer: 5 m

2 Two right-angled triangles

4. A radio tower PA is built in two sections: PB is the lower section and BA the upper one. Wires from a point O on the ground, 6 m from the foot P , hold the tower at B and at A . From O the angle of elevation of B is 30° and that of A is 60° . (i) Find the length of the wire OB . (ii) Find the length of the wire OA . (iii) Find AB , OR find the area of $\triangle OPB$. **CBSE 2026, 4 marks**



Angle $POB = 30^\circ$ and angle $POA = 60^\circ$

SOLUTION

(i) In $\triangle OPB$, $\cos 30^\circ = \frac{OP}{OB}$, so $OB = \frac{6}{\frac{\sqrt{3}}{2}} = \frac{12}{\sqrt{3}}$. $OB = 4\sqrt{3}$ m.

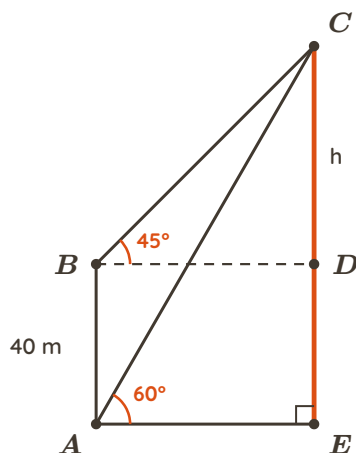
(ii) In $\triangle OPA$, $\cos 60^\circ = \frac{OP}{OA}$, so $OA = \frac{6}{\frac{1}{2}}$. $OA = 12$ m.

(iii) $PB = 6 \tan 30^\circ = \frac{6}{\sqrt{3}} = 2\sqrt{3}$ m and $PA = 6 \tan 60^\circ = 6\sqrt{3}$ m.

$AB = PA - PB = 6\sqrt{3} - 2\sqrt{3}$. $AB = 4\sqrt{3}$ m.

OR: area of $\triangle OPB = \frac{1}{2} \times OP \times PB = \frac{1}{2} \times 6 \times 2\sqrt{3}$. Area = $6\sqrt{3}$ m².

5. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60° . From a deck at B , 40 m directly above A , the angle of elevation of C is 45° . D is the point on CE level with B , and $CD = h$ metres. (i) Write BD in terms of h . (ii) Write BC in terms of h . (iii) Find the height CE , taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m. **CBSE 2025, 4 marks**



SOLUTION

(i) In $\triangle BDC$, $\tan 45^\circ = \frac{CD}{BD}$, so $BD = CD$. $BD = h$ metres.

(ii) $\sin 45^\circ = \frac{CD}{BC}$, so $BC = \frac{h}{\frac{1}{\sqrt{2}}}$. $BC = \sqrt{2} h$ metres.

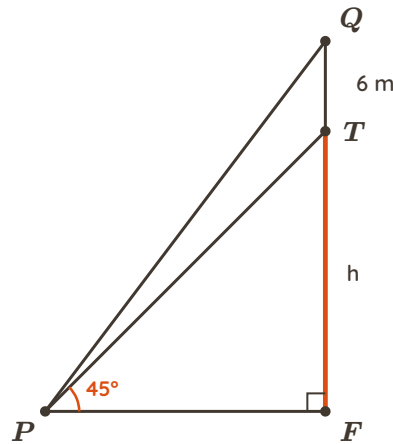
(iii) $AE = BD = h$ and $CE = h + 40$. In $\triangle AEC$, $\tan 60^\circ = \frac{h + 40}{h}$.

$$\sqrt{3} h = h + 40, \text{ so } h = \frac{40}{\sqrt{3} - 1} = \frac{40(\sqrt{3} + 1)}{2} = 20(\sqrt{3} + 1) = 20 \times 2.73 = 54.6$$

$CE = 54.6 + 40$. The lighthouse is 94.6 m high.

OR: in $\triangle AEC$, $\cos 60^\circ = \frac{AE}{AC}$, so $AE = 100 \times \frac{1}{2}$. $AE = 50$ m.

6. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.) **CBSE 2024, 5 marks**



FT is the tower and TQ is the pole

SOLUTION

Let the height of the tower be h m and the distance of P from its foot be d m.

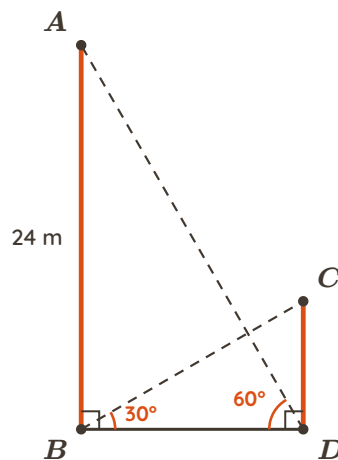
The angle of depression of P from the top of the tower is 45° , so the angle of elevation of the top of the tower from P is 45° ; $\tan 45^\circ = \frac{h}{d}$, so $d = h$.

For the top of the pole: $\tan 60^\circ = \frac{h+6}{d}$, so $\sqrt{3}h = h+6$.

$$h = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1) = 3 \times 2.73 = 8.19$$

Answer: the tower is 8.19 m high and P is 8.19 m from its foot.

7. Two towers stand on level ground. The first is 24 m high, and from the foot of the second tower its top is seen at an angle of elevation of 60° . From the foot of the first tower, the top of the second is seen at an angle of elevation of 30° . Find the distance between the towers, the height of the second tower, and the length of a wire joining the two tops. **CBSE 2023, 5 marks**



SOLUTION

Let $AB = 24$ m be the first tower, $CD = h$ the second, and $BD = d$ the distance between them.

In $\triangle ABD$: $\tan 60^\circ = \frac{24}{d}$, so $d = \frac{24}{\sqrt{3}} = 8\sqrt{3}$ m ≈ 13.86 m.

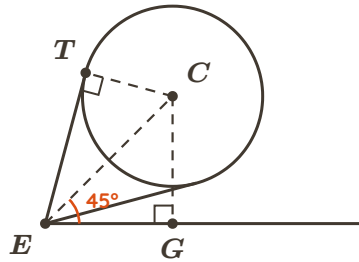
In $\triangle CDB$: $\tan 30^\circ = \frac{h}{d}$, so $h = \frac{8\sqrt{3}}{\sqrt{3}} = 8$ m.

The tops differ in height by $24 - 8 = 16$ m. Wire $AC = \sqrt{(8\sqrt{3})^2 + 16^2} = \sqrt{192 + 256} = \sqrt{448} = 8\sqrt{7}$ m.

Answer: distance $8\sqrt{3}$ m; second tower 8 m; wire $8\sqrt{7}$ m ≈ 21.17 m

8. A spherical balloon of radius r subtends an angle of 60° at the eye of an observer, and the angle of elevation of its centre from the eye is 45° . Prove that the centre of the balloon is at a height of $\sqrt{2}r$.

CBSE 2023, 5 marks



SOLUTION

Let E be the eye, C the centre of the balloon, ET a tangent to the balloon and CG the height of the centre above the level of the eye.

The two tangents from E make an angle of 60° , and EC bisects it, so $\angle CET = 30^\circ$.

$CT \perp ET$ (radius and tangent). In $\triangle CTE$: $\sin 30^\circ = \frac{r}{EC}$, so $EC = 2r$.

In $\triangle CGE$: $\sin 45^\circ = \frac{CG}{EC}$, so $CG = 2r \times \frac{1}{\sqrt{2}} = \sqrt{2}r$.

So the height of the centre is $\sqrt{2}$ times the radius. Proved.

9. A tower is 50 m high. From the foot of a building, the top of the tower is seen at an angle of elevation of 60° . From the foot of the tower, the top of the building is seen at an angle of elevation of 30° . Find the height of the building. CBSE 2022 Term 2, 3 marks

SOLUTION

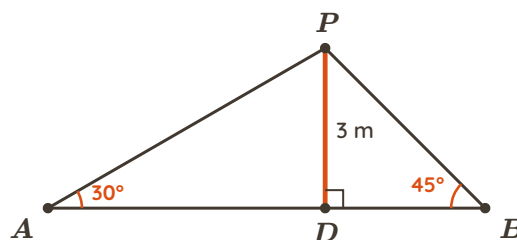
Let the distance between them be d and the height of the building be h .

$$\tan 60^\circ = \frac{50}{d}, \text{ so } d = \frac{50}{\sqrt{3}} \text{ m.}$$

$$\tan 30^\circ = \frac{h}{d}, \text{ so } h = \frac{50}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{50}{3}$$

Answer: $16\frac{2}{3}$ m, about 16.67 m

10. From a point on a bridge 3 m above the banks of a river, the two banks on opposite sides are seen at angles of depression of 30° and 45° . Find the width of the river. **CBSE 2022 Term 2, 3 marks**



SOLUTION

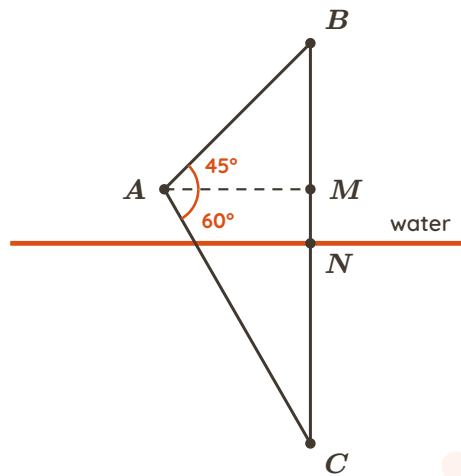
Let P be the point on the bridge, D the point directly below it, and A and B the two banks. The angles of depression equal the angles at A and B (alternate angles).

$$\tan 30^\circ = \frac{3}{AD}, \text{ so } AD = 3\sqrt{3} \text{ m. } \tan 45^\circ = \frac{3}{DB}, \text{ so } DB = 3 \text{ m.}$$

$$\text{Width} = AD + DB = 3\sqrt{3} + 3$$

Answer: $3(\sqrt{3} + 1)$ m, about 8.2 m

11. The top B of a pavilion standing in a lake is 10 m above the water. From a point A , h metres above the water, the angle of elevation of B is 45° and the angle of depression of the reflection C of B in the water is 60° . (i) Draw a labelled figure. (ii) Find h , taking $\sqrt{3} = 1.73$. **CBSE 2022 Term 2, 4 marks**



N is on the water surface, with $BN = NC = 10$ m

SOLUTION

(i) The reflection C is as far below the water surface as B is above it, so $NC = BN = 10$ m. Draw AM horizontal, meeting BC at M . Then $MN = h$.

(ii) $BM = 10 - h$ and $MC = 10 + h$.

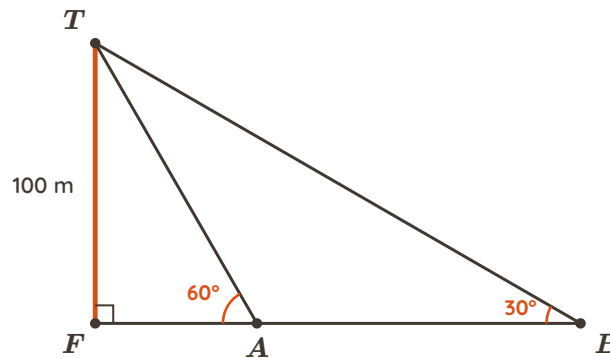
In $\triangle AMB$: $\tan 45^\circ = \frac{BM}{AM}$, so $AM = 10 - h$.

In $\triangle AMC$: $\tan 60^\circ = \frac{MC}{AM}$, so $\sqrt{3}(10 - h) = 10 + h$.

$$h(\sqrt{3} + 1) = 10(\sqrt{3} - 1), \text{ so } h = \frac{10(\sqrt{3} - 1)}{\sqrt{3} + 1} = \frac{10(\sqrt{3} - 1)^2}{2} = 5(4 - 2\sqrt{3}) = 20 - 10\sqrt{3}$$

Answer: $h = 20 - 17.3 = 2.7$ m

- 12.** A man rows a boat directly away from a lighthouse 100 m high. In 2 minutes the angle of elevation of the top of the lighthouse changes from 60° to 30° . Find the speed of the boat in metres per minute.
Take $\sqrt{3} = 1.732$. **CBSE 2019, 4 marks**



SOLUTION

Let TF be the lighthouse, and let the boat move from A to B .

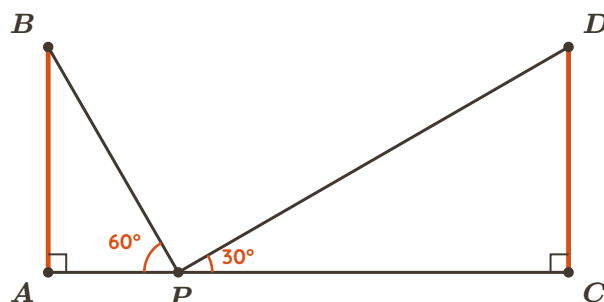
$$\tan 60^\circ = \frac{100}{FA}, \text{ so } FA = \frac{100}{\sqrt{3}} \text{ m. } \tan 30^\circ = \frac{100}{FB}, \text{ so } FB = 100\sqrt{3} \text{ m.}$$

$$AB = 100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{300 - 100}{\sqrt{3}} = \frac{200}{\sqrt{3}} = \frac{200\sqrt{3}}{3} \approx 115.47 \text{ m}$$

$$\text{Speed} = \frac{115.47}{2}$$

Answer: about 57.73 m per minute

- 13.** Two poles of equal height stand opposite each other on the two sides of a road 80 m wide. From a point on the road between them, the tops of the poles are seen at angles of elevation of 60° and 30° . Find the height of the poles and the distances of the point from them. **CBSE 2019, 4 marks**



SOLUTION

Let the poles AB and CD have height h , and let the point P be x m from the first pole, so it is $(80 - x)$ m from the second.

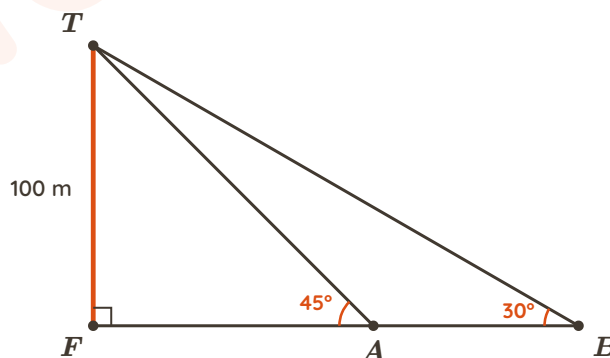
$$\tan 60^\circ = \frac{h}{x}, \text{ so } h = \sqrt{3}x. \tan 30^\circ = \frac{h}{80 - x}, \text{ so } h = \frac{80 - x}{\sqrt{3}}.$$

$$\sqrt{3}x = \frac{80 - x}{\sqrt{3}}, \text{ so } 3x = 80 - x \text{ and } x = 20.$$

$$h = 20\sqrt{3} \approx 34.64 \text{ m}$$

Answer: each pole is $20\sqrt{3}$ m high; the point is 20 m from one pole and 60 m from the other

- 14.** From the top of a lighthouse 100 m above sea level, two ships are seen at angles of depression of 30° and 45° . One ship is directly behind the other on the same side of the lighthouse. Find the distance between the ships. Take $\sqrt{3} = 1.732$. **CBSE 2018, 4 marks**



SOLUTION

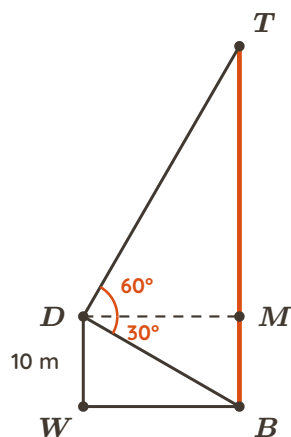
Let TF be the lighthouse and A, B the ships. The angles of depression equal the angles of elevation at the ships (alternate angles).

$$\tan 45^\circ = \frac{100}{FA}, \text{ so } FA = 100 \text{ m. } \tan 30^\circ = \frac{100}{FB}, \text{ so } FB = 100\sqrt{3} \text{ m.}$$

$$AB = 100\sqrt{3} - 100 = 100(1.732 - 1)$$

Answer: 73.2 m

15. A man stands on the deck of a ship, 10 m above the water. He sees the top of a hill at an angle of elevation of 60° and the foot of the hill at an angle of depression of 30° . Find the distance of the hill from the ship and the height of the hill. **CBSE 2016, 3 marks**



SOLUTION

Let D be the man, W the point on the water below him, BT the hill and DM the horizontal through D . Then $MB = DW = 10$ m.

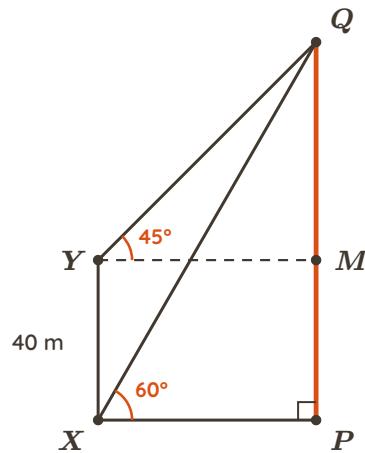
In $\triangle DMB$: $\tan 30^\circ = \frac{10}{DM}$, so $DM = 10\sqrt{3}$ m.

In $\triangle DMT$: $\tan 60^\circ = \frac{MT}{DM}$, so $MT = 10\sqrt{3} \times \sqrt{3} = 30$ m.

Height of the hill = $MT + MB = 30 + 10$

Answer: distance $10\sqrt{3}$ m (about 17.3 m); height 40 m

16. The top Q of a vertical tower PQ is seen at an angle of elevation of 60° from a point X on the ground. From a point Y , 40 m vertically above X , the angle of elevation of Q is 45° . Find the height PQ and the distance PX . Take $\sqrt{3} = 1.73$. **CBSE 2016, 4 marks**



SOLUTION

Draw YM horizontal, meeting PQ at M . Let $PX = YM = d$ and $PQ = h$, so $MQ = h - 40$.

In $\triangle YMQ$: $\tan 45^\circ = \frac{h - 40}{d}$, so $d = h - 40$.

In $\triangle XPQ$: $\tan 60^\circ = \frac{h}{d}$, so $h = \sqrt{3}d$.

$d = \sqrt{3}d - 40$, so $d = \frac{40}{\sqrt{3} - 1} = 20(\sqrt{3} + 1) = 20 \times 2.73 = 54.6$

$h = 54.6 + 40 = 94.6$

Answer: $PQ = 94.6$ m and $PX = 54.6$ m

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